

Wave Equation and its Solution

Wave equation is the most used second-degree partial differential equation in Applied Mathematics.

Examples: acoustic waves, water waves, electromagnetic waves and seismic waves.

- Acoustic waves are audible sound from a speaker (waves traveling through air at the speed of sound).
 - A seismic wave is a mechanical wave that travels through the Earth. It can result from an earthquake or volcanic eruption.
 - Electromagnetic waves are formed when an electric field couples with a magnetic field. Radio waves, ultraviolet and x-rays are examples of EM.
- ❖ To understand the wave equation, we are going to consider something simpler such as the mechanical vibration of an elastic string (violin and guitar string).
- ❖ The ability of an object or material to resume its normal shape after being stretched or compressed is known as elasticity of that object. Based on the material of the object, elasticity may differ from object to object.
- Suppose that an elastic string of length L is tightly stretched between two supports at the same horizontal level, so that the x -axis lies along the string. Suppose that the string is set in motion so that it vibrates in a vertical plane and let $u(x, t)$ denote the vertical displacement experienced by the string at the point x at time t . If damping effects, such as air resistance, are neglected, and if the amplitude of the motion is not too large, then $u(x, t)$ satisfies the partial differential equation

$$a^2 u_{xx} = u_{tt}$$

represents a wave equation.

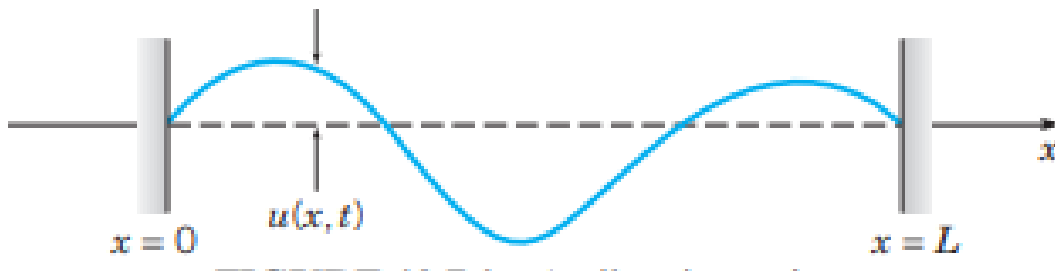
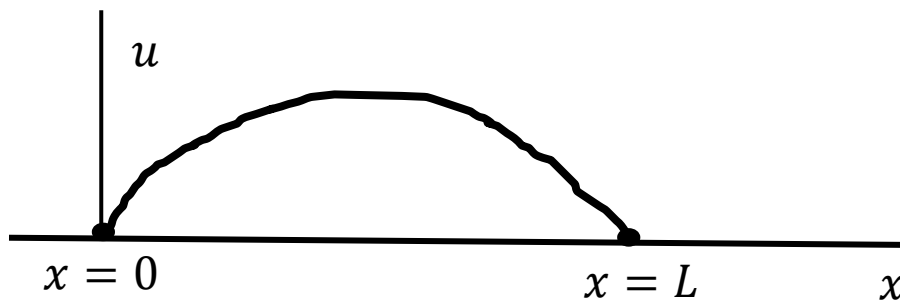


Figure: A vibrating String

Let $u(x, t)$ denote the vertical displacement of a string from x -axis at position x and time t . The string has length L . Its left- and right-hand ends are held fixed at height zero and we are told its initial configuration and speed. For notational convenience, choose a coordinate system so that the left-hand end of the string is at $x = 0$ and the right-hand end of the string is at $x = L$.

Problem: Determine $u(x, t)$.



We will consider the vibration of the string to have simple amplitude, so that it obeys the wave equation.

$$\frac{\partial^2}{\partial t^2} u(x, t) = a^2 \frac{\partial^2}{\partial x^2} u(x, t) \quad \text{for } 0 < x < L \text{ and } t > 0$$

Boundary Conditions: $u(0, t) = 0$ and $u(L, t) = 0$ for all $t > 0$

Initial Conditions: $u(x, 0) = f(x)$ and $u_t(x, 0) = g(x)$ for all $0 < x < L$

We need to determine $u(x, t)$ for all x and t .

Example-1: Determine $u(x, t)$ for the following Initial-Boundary value problem.

$$\frac{\partial^2}{\partial t^2} u(x, t) = a^2 \frac{\partial^2}{\partial x^2} u(x, t) \quad \text{for } 0 < x < 1 \text{ and } t > 0$$

Boundary Conditions: $u(0, t) = 0$ and $u(1, t) = 0$ for all $t > 0$

Initial Conditions: $u(x, 0) = x(1 - x)$ and $u_t(x, 0) = 0$ for all $0 < x < 1$

Example-2: Determine $u(x, t)$ for the following Initial-Boundary value problem.

$$\frac{\partial^2}{\partial t^2} u(x, t) = a^2 \frac{\partial^2}{\partial x^2} u(x, t) \quad \text{for } 0 < x < 1 \text{ and } t > 0$$

Boundary Conditions: $u(0, t) = 0$ and $u(1, t) = 0$ for all $t > 0$

Initial Conditions: $u(x, 0) = \sin(5\pi x) + 2 \sin(7\pi x)$ and $u_t(x, 0) = 0$ for all $0 < x < 1$